

FESHBACH METHOD IN APPLICATION TO THE GENERAL POLYNOMIAL POTENTIAL

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We consider the radial Schrödinger equation with an arbitrary superposition of the power-law potentials. In the (modified) harmonic-oscillator basis and for a sufficiently large model space, we construct the simple and closed effective hamiltonian matrix. Its asymptotically dominant component coincides with the recently suggested smooth-truncation prescription.

The Feshbach method of solving the Schrödinger equation [1]

$$H\psi = E\psi \quad (1)$$

is based on an introduction of projector P and rigorous conversion of (1) into an M -dimensional set of equations

$$\sum_{m=0}^{M-1} (H + H^\Delta)_{nm} \varphi_m = E \varphi_n, \quad n = 0, 1, \dots, M-1, \quad (2)$$

for $\varphi = P\psi$. By definition, we have

$$H^\Delta = H^\Delta(E) = -(H - E)Q[Q(H - E)Q]^{-1}Q(H - E),$$

$$Q = 1 - P, \quad (3)$$

so that the infinite-dimensional diagonalization (1) becomes replaced by the inversion (3). This is an important simplification, provided only that the E -dependence of $H^\Delta(E)$ is sufficiently smooth.

In the radial one-body form of (1)

$$[\delta + V(r)]\psi(r) = 0,$$

$$\delta = -d^2/dr^2 + l(l+1)/r^2 - E, \quad l = 0, 1, \dots, \quad (4)$$

we shall consider here any potential V defined as a superposition of rational powers $r^{m/n}$ of the coordinate. A change of variables $r \rightarrow r^{\text{const}}$ and $\psi \rightarrow r^{\text{const}}\psi$ makes such a potential equivalent to an even polynomial [2]. Moreover, the treatment of its singular $r^{-4}, \dots$ components does not change the essence of

our formalism [3]. Hence, we may work here with the simple anharmonic forces

$$V(r) = g_0 + g_1 r^2 + \dots + g_t r^{2t}, \quad g_t = 1, \quad t \geq 2, \quad (5)$$

without any loss of generality.

In the standard harmonic-oscillator (HO) basis $|n\rangle$ [4], the hamiltonian $H = \delta + E + V(r)$ will become a $(2t+1)$ -diagonal matrix [4]. Then, we are permitted to replace the operator $H - E$ in (3) by its Q_t -projection, ($Q_k = \sum_{m=M-i}^{\infty} |m\rangle\langle m|$, cf. ref. [3]) since $H^\Delta = RH^\Delta$, R is nonzero only for $R = PQ_t = Q_tP$. Next, we may write

$$Q_t(H - E)Q_t = \delta^{(\text{HO})} + \sum_{n=2}^t g_n Q_t r^{2n} Q_t = \delta(\beta, \gamma) + K,$$

$$\delta^{(\text{HO})} = Q_t(\delta + g_0 + g_1 r^2)Q_t = \delta(\beta, \gamma) + \beta + \gamma Q_t r^2 Q_t, \quad (6)$$

where β and γ are arbitrary parameters, and factorize

$$K = Q_t \left(\prod_{n=1}^t (r^2 - \alpha_n) \right) Q_t, \quad (7)$$

where, obviously, the new (complex) parameters $\alpha = \alpha_n(\beta, \gamma, g)$ are roots of the algebraic equation

$$\sum_{m=2}^t g_m \alpha^m + \gamma \alpha + \beta = 0. \quad (8)$$

Finally, we may introduce our main approximation formula

$$Q_t(H - E)Q_t = K [1 + O(1/M^{t-1})], \quad (9)$$

Table 1
The simplest anharmonic oscillators.

$V(r)$	r^2	r^3	r^4	r^5	r^6	r^7	r^8	r^9	r^{10}
complexity (ref. [5])	0	4	2	6	1	8	4	10	2
degree t [eq. (5)]	1	4	2	6	3	8	4	10	5
D is diagonal	no	yes	yes	yes	no	yes	yes	yes	no

which follows from the asymptotic estimate $\langle M+i | r^{2k} | M+j \rangle = O(M^k)$.

In our basis, the nonzero parts of the Q -projected operators may be treated as matrices. Thus, the matrix elements of the operators $Q_{t+i}(r^2 - \alpha) = Q_{t+i}(r^2 - \alpha)Q_{t+i+1}$ form the upper tridiagonal matrices $T_{t+1}(\alpha)$. We may define also the upper triangular (s -diagonal) matrices.

$$U = T_1(\alpha_1) T_2(\alpha_2) \dots T_s(\alpha_s), \quad (10)$$

and factorize, in an algebraically rigorous way,

$$K = U D L, \quad (11)$$

where the lower triangular (s -diagonal) matrix

$$L = T_s^T(\alpha_{s+j+1}) T_{s-1}^T(\alpha_{s+j+2}) \dots T_1^T(\alpha_{2s+j}). \quad (12)$$

is accompanied by the diagonal ($D=1$) or tridiagonal ($D = Q_{t+s}(r^2 - \alpha_{s+1})Q_{t+s}$) matrix D for $t = 2s+j$ with $j = 0$ or $j = 1$, respectively.

The main consequence of our approximation (9) and factorization (11) is the significant simplification of the corresponding effective hamiltonian (3). Indeed, when we employ the identities

$$R L = R L R, \quad U R = R U R$$

and

$$R K R - R K Q(Q K Q)^{-1} Q K R = R(R K^{-1} R)^{-1} R, \quad (13)$$

we get our main result

$$R(H + H^\Delta)R = R U R (R D^{-1} R)^{-1} R L R [1 + O(1/M^{t-1})]. \quad (14)$$

Thus, provided that the inverse of D is evaluated, eq. (2) becomes well defined. It may be solved on the computer, with the same precision $1 + O(1/M^{t-1})$.

Concerning the inversion D^{-1} , it is trivial ($D=1$) for both the odd and quartic-type oscillators r^{2s+1} and r^{4s} , respectively (cf. table 1). For the remaining "canonical" oscillators r^{4s+2} [5], we have to fix β or γ by the requirement

$$\alpha_{s+1}(\beta, \gamma, g) = 0 \quad (15)$$

(the solution of which exists, e.g., for $\pm\beta = \prod_{i=1}^t \alpha_i = 0$). Then, the inversion of $D = Q_{t+s} r^2 Q_{t+s}$ becomes also easy [into an analogue of eq. (13)]

$$(Q r^2 Q)^{-1} = r^{-2} - r^{-2} P(P r^{-2} P)^{-1} P r^{-2}, \quad (16)$$

it is sufficient to insert the matrix elements

$$\langle m | r^{-2} | n \rangle = (l + 1/2)^{-1} \left(\frac{n! \Gamma(m + l + 3/2)}{m! \Gamma(n + l + 3/2)} \right)^{1/2}, \quad m \leq n. \quad (17)$$

We may conclude that our construction is completed. The differential equation (4) + (5) has been reduced to the M -dimensional diagonalization (2) + (14). The precision of results may be controlled by our choice of M and it increases also with the growing parameter t in (5). In principle (especially for small t), it would be possible to incorporate also the $O(\delta(\beta, \gamma)/K)$ corrections by perturbative or variational techniques — this will not be done here.

References

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